

inputs: a vector
(petal width, sepal width,
both rescaled to 0 to 1)

weights: a matrix

biases: a vector

$$\begin{bmatrix} 0.5 & 0.4 \end{bmatrix}$$

$\times$

$$\begin{bmatrix} -2 & -1 \\ 1 & 0.5 \end{bmatrix}$$

$+$

$$\begin{bmatrix} 1.2 & 0.1 \end{bmatrix}$$

$=$

$$\begin{bmatrix} 0.6 & -0.2 \end{bmatrix}$$

first number: $0.5 \times (-2) + 0.4 \times 1 + 1.2 = 0.6$

second number: $0.5 \times (-1) + 0.4 \times 0.5 + 0.1 = -0.2$

ReLU: every negative
number becomes 0

$$\begin{bmatrix} 0.6 & 0 \end{bmatrix}$$

output of the layer,
the input of the next layer